Кр 2

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Група: Кт - 33

Варіант: 14

Хід роботи

$$F = (x_1 + 3x_2 + 2x_3 + 3x_4) \to max$$

$$x_1 + 2x_2 + x_3 + x_4 \leq 30$$

$$-x_1 + 3x_2 \leq 8$$

$$x_1 - 2x_2 \leq 2$$

$$x_3 + 2x_4 \leq 10$$

$$3x_3 + x_4 \leq 15$$

$$x_j \geq 0 \quad j = 1 \dots 4$$

1 Блок

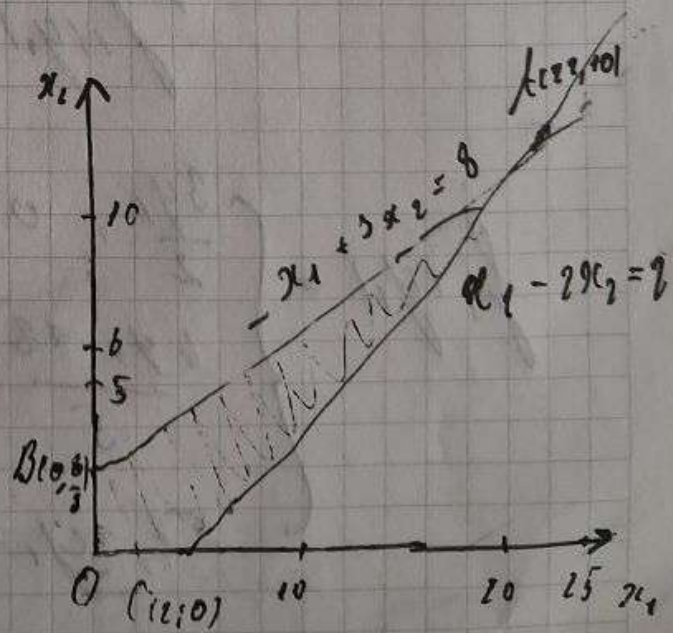
$$max(x_1 + 3x_2) = f_1(y_1)$$

$$x_1 + 2x_2 \leq y_1$$

$$-x_1 + 3x_2 \leq 8$$

$$x_1 - 2x_2 \leq 2$$

$$x_1 \geq 0, \; x_2 \geq 0, \; y_1 \geq 0$$



а) $0 < y_1 < \frac{16}{3}$

$x_1^* = 0$

$x_2^* = \frac{y_1}{2}$

$f_1(y_1) = \frac{3y_1}{2}$

б) $y_1 \geq \frac{16}{3}$

$-x_1 + 3x_2 = 8$

$x_1 + 2x_2 = y_1$

$x_1 = \frac{(3y_1 - 16)}{5}$

$x_2 = \frac{(y_1 + 8)}{5}$

$f_1(y_1) = \frac{(6y_1 + 8)}{5}$

Т. $k(22; 10)$ $y_1 = 42$

$f_1(y_1) = 52$

$$f_1(y_1) = \begin{cases} \frac{3y_1}{2}, & 0 < y_1 < \frac{16}{3} \\ \frac{6y_1 + 8}{5}, & \frac{16}{3} \leq y_1 \leq 42 \\ 52, & y_1 > 42 \end{cases}$$

Блок 2

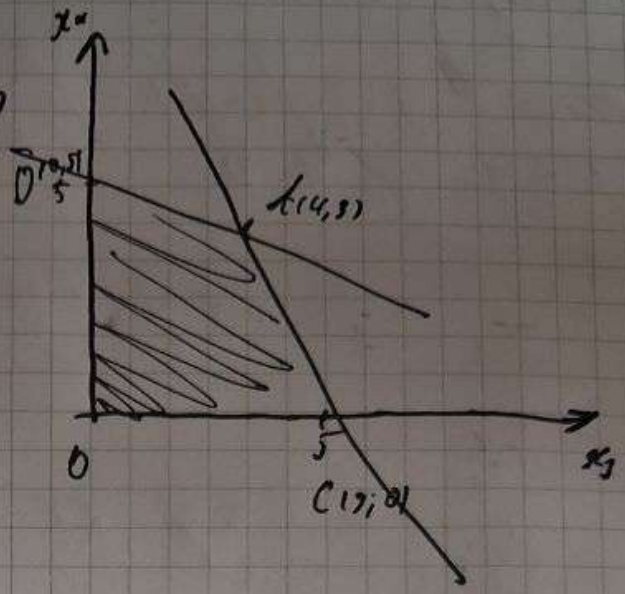
$$\max (2x_3 + 3x_4) = f_2(y_2)$$

$$x_3 + x_4 \le y_2$$

$$x_3 + 2x_4 \le 10$$

$$3x_3 + x_4 \le 15$$

$$x_3 \ge 0,\ x_4 \ge 0,\ y_2 > 0$$



а) $0 \le y_2 \le 5$

$x_3^0 = 0$

$x_4^0 = y_2$

$f_2(y_2) = 3y_2$

б) $y_2 > 5$

$x_3 + x_4 = y_2$

$x_3 + 2x_4 = 10$

$x_3 = 2y_2 - 10$

$x_4 = 10 - y_2$

$f_2(y_2) = y_2 + 10$

1. $A(4,3)$: $y_2 = 7$. $f_2(y_2) = 17$

$$f_2(y_2) = \begin{cases} 3y_2,\ 0 \le y_2 \le 5 \\ y_2 + 10,5 \le y_2 \le 7 \\ 17,\ y_2 > 7 \end{cases}$$

Координатна задача

max $F(y) = f_1(y_1) + f_2(y_2)$

$0 \le y_1 + y_2 \le 30$

1. $\frac{df_1}{dy_1} = \frac{3}{2}, \quad 0 \le y_1 \le \frac{16}{3}$

$\frac{df_2}{dy_2} = 3, \quad 0 \le y_2 \le 5$

$\frac{df_2}{dy_2} > \frac{df_1}{dy_1} \Rightarrow y_2 = 5$

2. $\frac{df_1}{dy_1} = \frac{3}{2}, \quad 0 \le y_1 \le \frac{16}{3}$

$\frac{df_2}{dy_2} = 1, \quad 5 \le y_2 \le 7$

$\frac{df_1}{dy_1} > \frac{df_2}{dy_2} \Rightarrow y_1 = \frac{16}{3}$

3. $\frac{df_1}{dy_1} = \frac{6}{5}, \quad \frac{16}{3} \le y_1 \le 42$

$\frac{df_2}{dy_2} = 1, \quad 5 \le y_2 \le 7$

$\frac{df_1}{dy_1} > \frac{df_2}{dy_2} \Rightarrow y_1 = 30 - 5 = 25$

$x_1^0 = \frac{13 \cdot 25 - 101}{5} = \frac{59}{5}$

$x_1^0 = \frac{25+8}{5} = \frac{33}{5}$

$x_1^0 = 0$

$x_4^0 = 5$

Перевірка:

$\frac{59}{5} + 2 \cdot \frac{33}{5} + 0 + 5 = 30$

$F_{max} = \frac{59}{5} + 3 \cdot \frac{33}{5} + 2 \cdot 0 + 3 \cdot 5 = 46,6$